

Rave Generation

PRESENTS



USER MANUAL

Introduction

Screech Machine is a three-oscillator synthesizer for screeches and leads. The oscillators feed a fixed chain of filter, drive, filter, waveshaper, OTT and EQ. The init patch is a plain saw with that chain wide open; the screeches are in the presets. A pitch envelope and five pitch contours shape how each note lands; a modulation envelope with eight depth pills, two LFOs and the mod wheel move the chain.

Signal Flow

OSC 1-3 → AMP → FILTER A → DRIVE → FILTER B → MORPH → OTT → EQ → PHASER → SAT → CHORUS → LEVEL → REVERB → CLIP

Everything from Filter A on is shared by all voices, so in Poly a chord distorts as one signal, like a chord played into a pedal.

Key Features

- 3 oscillators with 10 waves, including a 7-voice supersaw modeled on the JP-8000's, a PWM saw modeled on the Alpha Juno's and R-Mod (ring modulation)
- Hard sync of OSC 2 to OSC 1, swept by the pitch envelope, an LFO or the mod wheel
- Unison up to 9 voices per oscillator, with detune, stereo width and mix
- Pitch envelope of -48 to +48 semitones with five stackable contours: Drop, Dive, Rise, Bounce and Hoover
- Poly (16 voices), Mono and Legato, with glide
- A morphing LP / BP / HP filter with key and velocity tracking into four drive curves, then 11 filter models
- 256-curve wavetable waveshaper with wavefolding
- 3-band OTT compression, output EQ, a phaser and a last saturation, a chorus modeled on the JP-8000's and a stereo reverb
- Modulation envelope with eight depth pills, two LFOs with eleven destinations, mod wheel
- 1x, 2x or 4x oversampling
- The output never goes over 0 dBFS

User Interface Tips

Hint Bar

Hover over any control to read a one-line description in the bar at the bottom of the panel. The icon in the header switches the bar on and off.

Frequency Readouts

Filter A Freq, Filter B Freq and Peak Freq turn on a 0-10 scale; the readout under each shows the frequency in Hz.

Bar Switches

The OSC 2, PITCH, CHORUS and PHASER bars are switches: click one and its LED lights. OSC 2 switches sync, PITCH sends the pitch envelope to OSC 2 alone, CHORUS and PHASER switch their effects. PRE / POST at the left end of the CHORUS bar moves the chorus.

Section Colours

Red bars mark the sound path, grey the envelopes, teal unison and effects, purple the wheel and output.

Knobs

Hold Shift while dragging for fine adjustment. Right-click a knob to return it to its default (Alt+click works too, Option+click on macOS); double-click it to type a value.

Menu Bar

Undo and Redo; the panel's opacity, 0% to 100%, to see the DAW behind it; the interface size, 70% to 200%; and the preset menu, which browses, loads and saves presets and copies and pastes them between instances.

Oscillators

Three oscillators with the same four controls. In the init patch all three are saws, OSC 1 at full level and the other two at 0%.

Parameter	Range	Default	Description
Wave	10 waves	Saw	Oscillator waveform, see the list below
Semi	-24 to +24 st	0 st	Transpose in semitones
Fine	-50 to +50 ct	0 ct	Detune in cents
Level	0-100%	100 / 0 / 0%	Oscillator level

Waves

Wave	Description
Saw	Sawtooth
Super	7-voice supersaw modeled on the JP-8000's. Detune sets its spread, Width its stereo image and Mix the centre saw against the others; Voices does not apply
Square	Square
Pulse	25% pulse
Triangle	Triangle
Sine	Sine
Noise	White noise
PWM	A pulse whose width sweeps. PWM Rate and Depth set the sweep
Saw PWM	A saw plus a PWM pulse an octave up: the hoover source, modeled on the Alpha Juno's. PWM Rate and Depth set the sweep
R-Mod	The other two oscillators multiplied (ring modulation), at this oscillator's Level. Its Semi and Fine do nothing

Unison

Parameter	Range	Default	Description
Voices	1-9	1	Unison copies of every oscillator. Super ignores it
Detune	0-100%	30%	Spread of the copies, up to 100 cents either side. Also sets the Super detune
Width	0-100%	50%	Stereo spread of the copies and of the Super voices. 0 is mono
Mix	0-100%	100%	Level of the detuned copies against the centre one: 0 is the centre alone, 100 all equal. On Super, the centre saw against the others

The copies are scaled so the level stays about the same as voices or Mix change.

PWM

Parameter	Range	Default	Description
Rate	0-100%	40%	Sweep speed
Depth	0-100%	40%	How far the width sweeps. 0 is a plain square

Both affect only the PWM and Saw PWM waves.

Sync

Click the OSC 2 bar to switch sync on. OSC 2 then follows OSC 1's pitch, and its own tuning changes its tone. Set OSC 2 a few semitones above OSC 1 and move its pitch for the sync sweep: send the pitch envelope to OSC 2 (the PITCH bar), or put an LFO or the mod wheel on Osc 2 Pitch.

Every wave can take part except Noise and R-Mod.

Parameter	Range	Default	Description
Sync	Off, On	Off	Click the OSC 2 bar

Pitch, Amp and Voice

Pitch Envelope

An envelope on the pitch of all three oscillators, or of OSC 2 alone: up to Tune, held, then back to Sustain. Tune sets the distance, so at Tune 0 the envelope does nothing and only the Pitch Mode moves the pitch.

Parameter	Range	Default	Description
Tune	-48 to +48 st	0 st	How far the envelope moves the note
Attack	0-2000 ms	0 ms	Time to reach Tune
Curve	-100 to +100%	-50%	Shape of the attack and decay. Left snaps fast and then eases in, right starts slow
Hold	0-2000 ms	0 ms	Time at Tune before the decay
Decay	0-5000 ms	150 ms	Time from Tune back to Sustain
Sustain	0-100%	0%	Where the pitch rests while the key is held, as a share of Tune
Pitch Mode	6 options	Off	A fixed contour added on top of the envelope, see below
Dest	All, Osc 2	All	Click the PITCH bar: the envelope and Pitch Mode move OSC 2 alone, which sweeps a sync or an R-Mod

Pitch Modes

Each Pitch Mode adds a fixed contour to the envelope, so both run at once: a fast Drop can sit on a slow envelope rise. The contour restarts with every note, except overlapping notes in Legato.

Mode	Contour
Off	Envelope only
Drop	A fast drop into the note from an octave up. The standard screech attack
Dive	A slower dive into the note from two octaves up
Rise	A scoop up into the note from an octave below
Bounce	Drops from an octave up, overshoots the note and settles
Hoover	Swoops up from below, over the note and back, then sags slowly. Made for held notes

The Roar Vowel preset pairs the Hoover contour with the modulation envelope on Filter A Freq and Morph for the vowel movement.

Amp Envelope

Parameter	Range	Default	Description
Attack	0-5000 ms	2 ms	Rise time
Decay	0-10000 ms	300 ms	Fall time to Sustain
Sustain	0-100%	100%	Level while the key is held
Release	0-10000 ms	150 ms	Fade time after the key goes up

Voice

Parameter	Range	Default	Description
Voice Mode	Poly, Mono, Legato	Poly	Poly plays up to 16 voices. Mono retriggers and glides on every note. Legato glides between overlapping notes without retriggering
Glide	0-2000 ms	0 ms	Glide time, the same for any interval
Bend	0-36 st	2 st	Pitch bend range
Velocity			Moves Filter A's cutoff, through its Vel knob, and nothing else: the level does not follow it

The Chain

Six stages in series, shared by all voices: Filter A, Drive, Filter B, Morph, OTT and EQ, followed by the phaser, the saturation, the chorus, the output level, the reverb and a 0 dBFS clip.

Filter A

A filter that morphs from low-pass through band-pass to high-pass. It sits before the drive and picks the frequency that screams: a high Reso pushes a narrow band hard into the drive.

Parameter	Range	Default	Description
Freq	0-10	10	Cutoff, 15 Hz to 15.9 kHz
Reso	0-10	0	Resonance
Morph	-5 to +5	-5	Low-pass at -5, band-pass at 0, high-pass at +5
Key	0-100%	0%	How far Freq follows the played note. 100% tracks it one to one; in Poly it follows the latest note
Vel	-100 to +100%	0%	How far velocity moves Freq, up to 3 octaves either way. Above 0 harder notes open it, below 0 they close it

Drive

Parameter	Range	Default	Description
Type	Soft, Tube, Diode, Hard	Soft	Drive curve. Tube and Diode add even harmonics; Hard is a clipper
Drive	0-100%	0%	Amount of drive. 0 is clean
Saturation	0-100%	0%	In this panel, but it comes last in the chain: see Saturation

Filter B

Eleven filter models, after the drive.

Type	Description
LP Clean	12 dB/oct low-pass, clean
LP 3109 24	24 dB/oct low-pass modeled on the IR3109 filter chip
LP Ladder 24	24 dB/oct low-pass modeled on the Moog ladder
HP	12 dB/oct high-pass
BP	12 dB/oct band-pass
Peak	Resonant peak
Notch	Band reject
LP FM	Low-pass whose cutoff follows the incoming signal: the louder the drive hits it, the more it growls
Acid 24	24 dB/oct diode ladder modeled on the TB-303's
Ju-6	24 dB/oct low-pass modeled on the Juno-6's filter
Ju-106	The same, voiced as the Juno-106's, with softer resonance

Parameter	Range	Default	Description
Type	11 models	LP Clean	Filter model, see above

Parameter	Range	Default	Description
Freq	0-10	10	Cutoff, 21 Hz to 18.7 kHz (default fully open)
Reso	0-10	0	Resonance

Morph

Alien Drive's waveshaper: 256 curves in a 16 x 16 bank. X picks the curve family, from cleanest at 1 to most destructive at 16: folds, harmonic curves, clipping, rectifying and more. Y sets the hardness within the family.

Parameter	Range	Default	Description
X	1-16	1	Curve family
Y	1-16	4	Hardness within the family
Fold	0-100%	40%	How hard the signal runs through the curve. Past noon it folds back on itself
Mix	0-100%	0%	Shaper output against its own input

X and Y click through 16 steps. Heavy folding sounds cleanest at Quality 4x.

OTT

Parameter	Range	Default	Description
Depth	0-100%	0%	3-band upward and downward compression. 0 is off
Time	0-200%	100%	Attack and release together

Past about 50% Depth, OTT gets quieter and lets a short spike through at the start of each phrase.

EQ

Parameter	Range	Default	Description
Low Cut	20-1000 Hz	20 Hz	High-pass on the output. 20 Hz is off
Peak Freq	0-10	7.24	Centre of the peak band, 100 Hz to 12 kHz (default 3.2 kHz)
Peak Gain	-18 to +18 dB	0 dB	Boost or cut. 0 dB is off
Peak Q	0.3-10	2	Width of the band. Higher is narrower
High Cut	1-20 kHz	20 kHz	Low-pass on the output. 20 kHz is off

Phaser

Click the PHASER bar to switch it; off, the signal passes untouched.

Parameter	Range	Default	Description
Rate	0.05-10 Hz	0.5 Hz	Sweep speed
Depth	0-100%	70%	How far the sweep goes. Below 50% it also fades the phaser in; 0 is dry
Reso	0-100%	50%	Feedback: deeper notches and a ringing peak between them

Saturation

The last distortion in the chain, after the EQ and the phaser, so a swept EQ peak or the phaser's sweep gets distorted too. Its knob sits in the DRIVE panel; at 0 it is off.

Parameter	Range	Default	Description
Saturation	0-100%	0%	Amount of saturation. 0 is off

Chorus

A stereo chorus modeled on the JP-8000's. Click the CHORUS bar to switch it on. PRE / POST at the left end of the bar sets where it sits: POST at the end of the chain, PRE before the drive, so the drive and EQ work on the chorused sound.

Parameter	Range	Default	Description
On / Off	Off, On	Off	Click the CHORUS bar
Rate	0-100%	50%	Sweep speed
Depth	0-100%	50%	Sweep depth. Deeper is wider and wobblier. Below 50% it also fades the chorus in; 0 is dry
Position	Pre, Post	Post	Click PRE / POST at the left of the CHORUS bar. Pre: before the drive. Post: the end of the chain

Reverb

A stereo reverb at the end of the chain, after the LFO's Level and Pan, so a tremolo or a pan rings out in the tail.

Parameter	Range	Default	Description
Pre-Delay	0-250 ms	125 ms	Gap before the reverb; keeps the attack clear
Decay	0.01-20 s	6 s	Length of the tail
Mix	0-100%	0%	Dry against reverb. 0 is off, 100 the reverb alone

Output

Parameter	Range	Default	Description
Level	-24 to +12 dB	0 dB	Output level. The output never goes over 0 dBFS
Quality	1x, 2x, 4x	2x	Oversampling for the drive and the shapers: higher is cleaner on heavy drive and folding. In the header, next to the i icon

Modulation

The modulation envelope, two LFOs and the mod wheel move the chain. On a shared target their amounts add up.

Depth Scale

Every source uses the same scale: 100% moves a target over its knob's full travel, and Morph X and Y through all 15 steps.

Mod Env

One envelope for the whole chain. It restarts on every note (in Legato only after a gap) and releases when the last key goes up.

Parameter	Range	Default	Description
Attack	0-5000 ms	2 ms	Rise time
Decay	0-10000 ms	400 ms	Fall time to Sustain
Sustain	0-100%	0%	Level while a key is held
Release	0-10000 ms	200 ms	Fall time after the last key goes up

Depth pills: one under each of Filter A Freq, Reso and Morph, Drive, Filter B Freq, Morph X, Morph Y and Peak Freq (-100 to +100%, default 0%). Drag a pill to set how far the envelope moves that knob; the ring around the knob shows the span.

LFO 1 and LFO 2

Parameter	Range	Default	Description
Dest	12 options	Pitch / A Freq	What the LFO moves: Off, Pitch, A Freq, A Morph, Drive, B Freq, Morph X, Morph Y, Peak Freq, Level, Pan or Osc 2 Pitch (OSC 2 alone, as Pitch)
Shape	7 shapes	Sine / Triangle	Sine, Triangle, Saw Up, Saw Down, Square, S&H (steps) or Random (glides)
Mode	Free, Sync, Trig	Free / Sync	Free runs in hertz, Sync locks to the host tempo, Trig restarts the cycle on each note
Rate	0.05-40 Hz	5.5 / 2 Hz	Speed in Free and Trig. A 20 Hz pitch LFO gives a metallic shimmer
Rate (Sync)	2/1 to 1/32	1/8 / 1/8	Note division in Sync, including triplets. It takes the place of the Hz knob
Depth	-100 to +100%	0%	How far the LFO moves its target
Fade	0-5000 ms	0 ms	The depth ramps in over this time from each note, for delayed vibrato

Mod Wheel

Parameter	Range	Default	Description
Dest	12 options	Off	What the wheel moves, from the LFO list
Depth	-100 to +100%	100%	How far the wheel at the top moves its target. At rest it adds nothing

Installation

System Requirements

- Windows 10/11 (64-bit) or macOS 10.13+
- VST3 or AU: the AU version for Logic Pro, the VST3 version for every other DAW

Support

For technical support, updates and additional information:

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